Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Syrine Ben Aziza	Email	Syrine.benaziza@tno.nl
Institute Name	TNO			
Address	Anna van Buerenplein 1 2595 DA Den Haag The Netherlands			

2. Technology Overview

Title of Proposed Technology	Predictive Risk Monitoring for Safe and Reliable AI Systems			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity (√), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information / Communication		Knowledge Services	
	Medium Category	Small Category	Medium Category	Small Category
	Software	Software solutions	R&D / Engineering services	Other R&D / Engineering-related Technologies

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Concept of Proposed Technology	AI systems are increasingly deployed in operational environments where reliability, predictability, and trust are critical to commercial success. In practice, failures rarely occur abruptly; instead, system behavior degrades gradually due to data drift, changing operating conditions, software updates, hardware ageing, or system interactions. Such effects are difficult to detect with conventional monitoring and pose significant risks for SMEs deploying AI-enabled products. The proposed technology addresses this challenge through predictive risk monitoring for operational AI systems. Rather than reacting to failures after they occur, it anticipates emerging risks by analysing operational data over time and identifying early indicators of degradation or abnormal behavior. Core functions include continuous analysis of operational and performance data, early detection of drift or emerging risks, and generation of interpretable, actionable risk signals for engineers and operators. The technology integrates into existing AI pipelines, enabling scalable deployment as systems evolve, are updated, or are redeployed in new contexts. Unlike approaches focused primarily on algorithmic performance, it targets real-world industrial and regulated environments. The approach builds on TNO's experience with AI deployment in non-stationary settings, where reliability, interpretability, and lifecycle management remain persistent challenges. This includes work on predictive maintenance, which revealed limitations of purely data-driven methods and the importance of system integration and explainability (1), as well as research on digital twins for cyber-physical energy systems, highlighting the need for monitoring and adaptation over time (2)–(3). More recent work on reliability-by-design for digital twin–enabled systems address trust and reliability across the full system lifecycle (4). In parallel, TNO has developed software architectures for adaptive and scalable data processing, supporting evolutionary change in distributed analytics and streaming systems (5)–(6). These insights have been applied in large-scale projects across the energy, mobility, and infrastructure domains, where AI components operate within complex, continually changing systems. Examples include European and national projects such as ZEFES[1] (zero-emission long-haul transport), ReliaBlade2-NL[2] (structural health monitoring and lifetime prediction of wind turbine blades using digital twins), ENERSHARE[3] (secure and trustworthy energy data spaces), MAGPIE[4] (digitalisation and

[1] ZEFES \| Taking Zero-Emission Long-Haul Freight Transport in Europe to the next level

[2] grow-offshorewind.nl/project/reliablade2-nl

[3] https://enershare.eu/

[4] MAGPIE - European project smart green ports

	automation for sustainable port operations), and the AI-hub Noord-Nederland[5], where TNO collaborates with industrial partners, SMEs, and public stakeholders on responsible and trustworthy AI deployment. This background positions TNO to address predictive risk monitoring as a system-level design challenge rather than a standalone algorithmic problem. The proposed R&D builds on these lessons while remaining exploratory and feasibility-focused, recognising the need for iterative validation and close collaboration with industrial partners.
Rationale for Joint R&D Project	Most industrial AI systems rely on static models and threshold-based monitoring, which are inadequate for anticipating gradual degradation or complex failure modes under real operational conditions. As AI systems are increasingly expected to evolve over time, monitoring approaches must remain effective under non-stationary conditions rather than relying on assumptions made at deployment. Developing predictive risk monitoring therefore requires targeted R&D that combines time-series modelling, uncertainty estimation, and system-level analysis within an interpretable framework. Existing solutions rarely integrate risk prediction with operational decision support, which limits adoption by SMEs. SMEs typically lack the resources to develop such interdisciplinary capabilities independently. In this complex and dynamic landscape, TNO contributes expertise in predictive modelling, system reliability, and trustworthy AI, developed through applied projects in industrial domains. However, feasibility and robustness can only be validated through close collaboration with industrial partners operating AI systems in real-world settings. Korean SMEs provide access to operational AI systems, deployment environments, and usage patterns that differ from European contexts in both scale and application. Joint R&D within the TIPA framework enables validation across diverse conditions and supports the development of predictive monitoring approaches that are scalable across systems, deployments, and markets, strengthening global competitiveness.
Key Words	1. Predictive Risk Monitoring 2. Safe and Reliable AI 3. AI System Degradation

[5] Home - AI-hub Noord

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	4. Operational AI Monitoring 5. Trustworthy AI

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 3	End of the Project	Level 4
R&D Objectives	**Phase 1 Objectives (Month 1–6)** – Assess feasibility of predictive risk monitoring approaches for early detection of AI system degradation and emerging risks in operational settings. – Identify operationally meaningful and interpretable risk indicators using SME-provided use cases – Identify architectural options, technical constraints, and design trade-offs for integrating predictive risk monitoring into existing AI systems and pipeline – Define clear requirements and scope for Phase 2 **Phase 2 Objectives (Month 7–36)** Building on Phase-1 results, Phase 2 aims to: – Implement and validate predictive risk monitoring solutions in operational or pilot SME AI environments. – Demonstrate robustness, scalability, and reliability of monitoring approaches across evolving operating conditions, system updates, or deployments. – Support integration of predictive risk monitoring into SME products, services, or internal operational workflows.			
R&D Target Specifications	**Phase 1 – Targets** – Demonstrate that predictive monitoring can provide earlier or richer insight into AI system degradation or emerging risks compared to conventional reactive monitoring, using prototype-level analysis on historical or controlled operational data. – Identify a set of use-case-specific and interpretable risk indicators that show a meaningful relationship with observed degradation or abnormal system behavior. – Provide indicative evidence of earlier detection of degradation patterns compared to baseline monitoring (e.g. on the order of 10–20%, depending on the use case and data availability). – Demonstrate prototype-level integration of predictive risk monitoring within an existing AI pipeline or system architecture for one representative SME application. **Phase 2 – Targets** Phase 2 targets focus on validation and scaling:			

	– Operationally validated predictive risk monitoring deployed in pilot or operational SME AI environments – Demonstrated robustness and reliability of monitoring under changing operating conditions, system updates, and data drift – Actionable and interpretable risk signals supporting engineering and operational decision-making – Reusable monitoring architectures, deployment guidelines, and design patterns supporting broader SME adoption
R&D Plan	The project is structured in two phases, **Phase 1 (6 months)** and **Phase 2 (approximately 30 months)**. This phased approach ensures early insight with limited risk, followed by focused development once feasibility and value have been demonstrated. **Phase 1 – Feasibility and Design Definition (Month 1–6)** **Objective:** Establish feasibility, define architectural options, and scope Phase 2. **Key Tasks and Timeline** – **Task 1 – Partner alignment and baseline definition (Month 1–2)** Define the representative operational AI system and monitoring context; document baseline system behaviour and existing reactive monitoring practices, including data availability and operational constraints. – **Task 2 – Exploratory design of predictive monitoring concepts (Month 2–3)** Identify and compare candidate modeling and analysis approaches for predictive risk monitoring; define use-case-aligned risk indicators and high-level monitoring architecture options. – **Task 3 – Prototype-level feasibility exploration (Month 3–5)** Implement limited prototype components to explore feasibility of selected predictive monitoring approaches and perform exploratory evaluation using historical or controlled operational data. – **Task 4 – Consolidation and Phase-2 scoping (Month 5–6)** Consolidate findings; assess technical, organizational, and scalability challenges; and define objectives, success criteria, and resource requirements for Phase 2. **Milestones (Phase 1)** **M1:** Partner alignment completed and baseline monitoring context documented **M2:** Feasibility of predictive risk monitoring concepts evaluated **M3:** Evaluation report and Phase-2 roadmap delivered **Phase 2 – Development, Validation, and Scaling (Month 7–36)** **Objective:** Develop, validate, and demonstrate predictive risk monitoring for AI systems in

operational SME environments.

Key Tasks and Timeline

- **Task 5 – Implementation of predictive monitoring components (Month 7–14)**
  Implement and refine predictive risk monitoring components based on Phase-1 design choices and integrate them into selected SME AI systems or pipelines.
- **Task 6 – Validation under operational variability (Month 12–20)**
  Validate predictive monitoring approaches under changing operating conditions, system updates, and data drift, assessing robustness and early-warning capability.
- **Task 7 – Integration into SME workflows (Month 18–26)**
  Integrate predictive risk signals into SME operational workflows, dashboards, or decision-support processes, focusing on interpretability and usability for engineers and operators.
- **Task 8 – Performance and impact evaluation (Month 24–32)**
  Evaluate monitoring performance, robustness, and operational impact under realistic conditions, including earlier detection of degradation and improved reliability.
- **Task 9 – Consolidation, valorisation, and guidance (Month 30–36)**
  Consolidate results into reusable monitoring architectures, design guidelines, and best practices, and support valorisation and adoption aligned with SME business models.

Milestones (Phase 2)

M4: Predictive monitoring components implemented in SME environment
M5: Monitoring performance validated under operational variability
M6: Operational impact and reliability benefits evaluated
M7: Scalable reference architecture and adoption guidelines delivered

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	TNO will act as the **technical and methodological lead**, responsible for the design, feasibility assessment, and validation of predictive risk monitoring concepts for operational AI systems, in close collaboration with the SME partner. **In Phase 1, TNO will:** – Design and assess predictive risk monitoring concepts and system architectures tailored to the SME's AI application and operational context. – Identify, select, and evaluate candidate modeling and analysis approaches for early detection of AI system degradation or emerging risks, focusing on feasibility, interpretability, and integration effort. – Perform system-level analysis to identify technical assumptions, constraints, and design trade-offs related to monitoring accuracy, robustness under non-stationary conditions, and operational usability. – Consolidate findings into clear design recommendations and feasibility conclusions, including identified risks and priorities, to support decision-making and define a realistic scope for Phase 2. **In Phase 2, TNO will:** – Support the implementation, validation, and refinement of selected predictive risk monitoring approaches based on Phase-1 outcomes. – Contribute to performance, robustness, and scalability evaluation in operational or pilot SME environments, including assessment under changing operating conditions or system updates. – Support the development of reusable reference architectures, design guidelines, and technical documentation to facilitate adoption, scaling, and potential commercialization.
Expected R&R of Korean SME Partner	The Korean SME will act as the **industrial application and validation partner**, ensuring that the project remains grounded in real operational, technical, and business needs. **In Phase 1, the Korean SME will:** – Provide a representative AI application context, including relevant information on system behavior, operational workflows, data characteristics, and known reliability or monitoring challenges. – Participate actively in feasibility validation activities, including review of intermediate results and feedback on interpretability, practical relevance, and integration effort. – Contribute to the joint assessment of business relevance and valorisation potential, helping to identify success criteria and priorities for a potential Phase 2. **In Phase 2, the Korean SME will:**

	– Support pilot deployments and operational validation of predictive risk monitoring within relevant systems or environments. – Contribute to evaluation of monitoring performance, early-warning capability, and operational impact under realistic conditions. – Explore pathways for integration of the project outcomes into products, services, or internal workflows, supporting future commercialization or internal adoption.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Expected Outcomes – Phase 1 (Feasibility Phase)** At the end of Phase 1, the project is expected to deliver: – A feasibility-validated concept for predictive risk monitoring of operational AI systems, suitable for integration into existing SME AI pipelines. – A validated indication of early-warning potential, demonstrating how predictive monitoring can anticipate system degradation or emerging risks compared to conventional reactive monitoring, explored in a controlled or representative SME context. – Indicative evidence of added operational insight, such as earlier detection of degradation patterns or richer understanding of system behaviour, based on exploratory evaluation rather than full operational deployment. – Architectural design guidance and technical recommendations, including identified assumptions, constraints, risks, and priorities for further development and integration. **Expected Outcomes – Phase 2 (Development and Scaling Phase)** Building on the results of Phase 1, Phase 2 is expected to deliver: – Implemented and validated predictive risk monitoring components, operating in pilot or operational SME AI environments. – Operational validation of predictive monitoring approaches, demonstrating early detection of degradation, drift, or emerging risks under realistic and evolving conditions. – Measured improvements in operational reliability, such as earlier intervention, reduced unplanned downtime, or improved confidence in AI system behaviour, compared to conventional monitoring practices. – Reusable reference architectures, design patterns, and technical guidelines supporting adoption of predictive risk monitoring across SME products, services, or deployments. These outcomes are intended to support both technical scaling and practical integration of predictive risk monitoring into SME operations.

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	For SMEs, the project enables: – Earlier awareness of AI system degradation and emerging risks, supporting proactive intervention rather than reactive response. – Reduced operational risk and improved reliability of AI-enabled systems deployed in real-world environments. – Increased trust and confidence in AI system behaviour among engineers, operators, and stakeholders. **From a business perspective, these improvements support:** – Reduced costs associated with downtime, failures, and emergency interventions. – More predictable operation and lifecycle management of AI-enabled products and services. – Clearer pathways from experimental deployments to reliable, scalable offerings, reducing the risk of stalled adoption. **From an international collaboration perspective:** – Korean SMEs benefit from early validation of predictive risk monitoring approaches with a European research partner. – European partners gain exposure to diverse operational AI environments and usage patterns in the Korean SME ecosystem. – Jointly developed monitoring approaches are validated across heterogeneous conditions, strengthening applicability in international markets. Overall, the project contributes to safer, more reliable, and more trustworthy adoption of AI technologies by SMEs, strengthening competitiveness while reducing technical and operational risk.

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Titles;
(TNO) Paolo Pileggi: Dr.
(TNO) Elena Lazovik: MSc
(former TNO) Toon Albers: Msc
(TNO) Jacques Verriet: Dr
(TNO) Jeroen Broekhuijsen: Msc
(TNO) Michael Borth: Dr.
(TNO) Bram van der Waaij: Msc
(TNO) Robert J. Meijer: Prof. Dr.
(TNO) Wilco Wijbrandi: Msc